

# SAI VENKATA NAGENDRA ABHAY RAJ GURRAM

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## PROFESSIONAL SUMMARY

Generative AI and Machine Learning Engineer graduate student with hands-on experience building LLM-powered applications, RAG systems, and production-ready AI workflows. Strong proficiency in Python and transformer-based architectures, with practical experience fine-tuning deep learning models, designing data pipelines, and deploying scalable AI solutions in cloud environments. Experienced in optimizing neural networks for NLP tasks and implementing validation-driven model evaluation.

## TECHNICAL SKILLS

Programming: Python (Advanced), SQL

Generative AI & LLMs: GPT-style Models, Transformer Architectures, RAG Pipelines, Model Fine-Tuning, Prompt Engineering

Machine Learning: Deep Neural Networks, CNN (Foundational), NLP, Model Evaluation, Cross-Validation

AI Engineering: End-to-End ML Workflows, Data Pipeline Design, Data Preprocessing, Feature Engineering

Cloud & Deployment: AWS SageMaker, GCP Cloud Run, REST APIs, Production Deployment Concepts

Data & Tools: Pandas, NumPy, SciPy, scikit-learn, PyTorch, Streamlit, Git

Concepts: Neural Network Optimization, Model Distillation (Conceptual), AI Validation & Guardrails (Foundational), OOSD

## GENERATIVE AI & MACHINE LEARNING PROJECT EXPERIENCE

- 1 LLM-Powered RAG Application — Designed and implemented Retrieval-Augmented Generation pipeline integrating large language models for contextualized response generation; built scalable data ingestion and validation workflows.
- 2 Transformer-Based Sentiment Modeling — Fine-tuned BERT (PyTorch) for large-scale NLP classification; optimized training workflow, evaluated performance metrics, and structured deployment-ready outputs.
- 3 AI Workflow Automation Tool — Developed modular Python system integrating multi-source APIs; automated data preprocessing, feature extraction, and model evaluation processes to support scalable ML experimentation.

## INDUSTRY EXPERIENCE

Laurus Labs Ltd. — Quality Assurance Intern (Jan 2022 – June 2022)

- Developed SQL-based validation scripts to maintain data integrity across enterprise systems
- Collaborated with cross-functional teams to support automation and workflow optimization
- Documented processes ensuring reliability and scalability of operational data pipelines

## EDUCATION

Stevens Institute of Technology — M.S. Business Analytics & Artificial Intelligence (Expected May 2026)